

IP

IP Address Notes (Detailed)

1. Definition of IP Address

IP = Internet Protocol

An IP Address is a unique address given to a device connected to a network.

It helps devices identify each other and communicate.

Just like a house address helps letters reach the correct house, an IP address helps data reach the correct device.

Examples of devices with IP:

- Laptop
- Mobile phone
- Router
- Printer
- Server
- CCTV
- Smart TV
- Virtual Machine

Example IP:

192.168.1.10

2. Why IP is Needed

Without IP:

- no Internet
- no browsing
- no gaming
- no server communication
- no networking

- no cybersecurity scanning

IP helps data know:

- where to go
 - where it came from
 - how to return
-

3. Types of IP Address

Mainly:

IPv4

and

IPv6

4. IPv4

Most common version.

Format:

192.168.1.1

Rules:

- 4 parts (octets)
- separated by dots
- each part = 0 to 255
- total = 32 bits

Example:

8.8.8.8

Problem:

Limited addresses (~4.3 billion)

5. IPv6

Created because IPv4 addresses are limited.

Format:

2001:0db8:85a3:0000:0000:8a2e:0370:7334

Rules:

- hexadecimal
- separated by colons
- total = 128 bits

Very huge number of addresses.

6. Private IP Address

Used inside local network only.

Not directly accessible from the Internet.

Official private ranges:

10.0.0.0/8

172.16.0.0/12

192.168.0.0/16

Examples:

- 10.1.1.1
- 172.20.10.5
- 192.168.1.1

Used in:

- home Wi-Fi
- offices
- labs
- virtual machines

6.2. CIDR with Only 0-based Common Masks

CIDR

Full Form: Classless Inter-Domain Routing

Definition:

CIDR is a method used to show how many bits of an IP address are used for the **network part** and how many are used for the **host part**.

- **/8** → 255.0.0.0
First 8 bits = Network, remaining = Hosts
- **/16** → 255.255.0.0
First 16 bits = Network, remaining = Hosts
- **/24** → 255.255.255.0
First 24 bits = Network, remaining = Hosts

6.3. Private IP Formats + CIDR

CIDR

Full Form: Classless Inter-Domain Routing

Definition:

CIDR is used to show how many bits of an IP address are used for the **network part** and how many are used for the **host part**.

Private IP Formats**a. Class A Private IP****Format:**

10.x.x.x

Common CIDR:

- **/8** → 255.0.0.0
-

b. Class B Private IP**Format:**

172.16.x.x \text{ to } 172.31.x.x

Common CIDR:

- **/12** → 255.240.0.0
-

c. Class C Private IP

Format:

192.168.x.x

Common CIDR:

- /16 → 255.255.0.0
 - Common home use: /24 → 255.255.255.0
-

7. Public IP Address

Visible on the Internet.

Assigned by ISP like:

- Jio
- Airtel
- BSNL

Examples:

- 8.8.8.8
- 1.1.1.1
- 49.x.x.x

Public IP has no fixed starting format.

It is basically any IP that is not private or reserved.

8. Static IP

Fixed IP address.

Does not change.

Used for:

- servers
- CCTV
- printers
- hosting

Example:

192.168.1.50

9. Dynamic IP

Changes automatically.

Assigned by DHCP.

Used by:

- phones
- laptops
- home Wi-Fi users

Most users use dynamic IP.

10. Loopback IP

Used when a computer talks to itself.

Main loopback IP:

127.0.0.1

Called:

localhost

Used for:

- testing
- local servers
- development

Example:

ping 127.0.0.1

Full loopback range:

127.0.0.0/8

11. APIPA

Automatic Private IP Addressing

Example:

169.254.x.x

Appears when DHCP fails.

Means system could not get proper IP from router.

12. Subnet Mask

Used to divide:

Network Part

and

Host Part

Example:

IP:

192.168.1.10

Mask:

255.255.255.0

This means:

- Network = 192.168.1.0
 - Host = 10
-

13. Gateway

Usually the router IP.

Acts like the exit door to the Internet.

Example:

192.168.1.1

Without gateway:

LAN works

but Internet access may not work.

14. DNS

Domain Name System

Converts:

google.com

into IP address.

Examples:

- Google DNS = 8.8.8.8
- Cloudflare DNS = 1.1.1.1

15. DHCP

Dynamic Host Configuration Protocol

Automatically gives private IP addresses.

Usually handled by router.

Example:

Device joins Wi-Fi

↓

Router gives 192.168.1.10

16. NAT

Network Address Translation

Converts:

Private IP → Public IP

Example:

192.168.1.5

↓

49.x.x.x

Allows many devices to share one public IP.

17. Important Commands

Windows

ipconfig

Shows:

- IP
- subnet mask
- gateway
- DNS

ping google.com

Tests connectivity

tracert google.com

Shows route

Linux

ip addr

ifconfig

ping

traceroute

18. Example from Your System

Your active Wi-Fi IP:

10.136.224.18

Gateway:

10.136.224.140

Subnet Mask:

255.255.255.0

These are private IPs.

19. Cybersecurity Use

Very important for:

- scanning
- enumeration
- recon
- exploitation

Example:

`nmap 192.168.1.0/24`

Without IP understanding, tools like Nmap become difficult.

Final Definition

IP Address = Network Identity of a Device

It helps communication happen between devices across networks.